

DSAA3073: Theories in Data Science

Week 1 Open Investigation

Probability Guarantees When Observations Depend

From a monitoring decision to a general analysis of assumptions

Basic analysis. A regional payment-monitoring case

Use only the probability tools developed in the Week 1 Learning Sheet

At 10:02, the automated monitor for the QuickCart checkout service reports an anomalous result in the Guangzhou region. During the preceding two-minute interval, it issued $n = 200$ synthetic payment requests. Of these, 40 terminated by timeout. Historical calibration under normal operation assigns timeout probability $p_0 = 0.05$ to an individual request.

Delaying intervention during a genuine incident may prevent customers from completing payment. Escalating a false alert interrupts the engineering team during a release. All 200 requests concern the same service, region, and short time interval. The monitor reports only end-to-end outcomes.

For the basic analysis, adopt the provisional model

$$X_1, \dots, X_{200} \text{ are i.i.d. Bernoulli}(0.05),$$

where $X_i = 1$ denotes a timeout and $X_i = 0$ denotes success.

Basic Task 1. Translate the case into a probability statement

Define the empirical timeout rate $\hat{p}$. Evaluate it from the recorded data and express the event of interest as a deviation from p_0 . State clearly which quantity is observed and which probability is supplied by the model.

Working space

Basic analysis. Apply the Week 1 guarantees

Mean information, variance information, and bounded independence

Basic Task 2. Derive and compare three upper bounds

Using the provisional i.i.d. model, carry out the following analysis.

- Apply Markov's inequality directly to the nonnegative statistic $\hat{p}$.
- Compute $\text{Var}(\hat{p})$ and apply Chebyshev's inequality to $|\hat{p} - 0.05| \geq 0.15$.
- Apply the two-sided Hoeffding inequality for independent variables in $[0, 1]$ with $n = 200$ and $\varepsilon = 0.15$.

Report all three numerical bounds. For each derivation, identify the exact information or assumption that makes it valid.

Working space

Basic Task 3. Make the operational decision

On the basis of the provisional model and the asymmetric costs, state an immediate action. Then state the strongest statistical conclusion that is justified at this stage. Write the conclusion conditionally, so that its dependence on the model is explicit.

Working space

Thinking investigation. Formalize the assumption-guarantee relation

The statistic may remain unchanged while the valid conclusion changes

Write a statistical investigation as the quadruple

$$(\mathcal{A}, \theta, \hat{\theta}, \mathcal{G}),$$

where $\mathcal{A}$ specifies the joint data-generating assumptions, θ is the estimand, $\hat{\theta}$ is the statistic, and $\mathcal{G}$ is the guarantee that the assumptions justify. In the basic analysis,

$$\mathcal{A}_0 : X_i \text{ are i.i.d. Bernoulli}(p_0), \quad \theta = p_0, \quad \hat{\theta} = \hat{p}.$$

The next three investigations modify $\mathcal{A}$ while keeping the observed statistic fixed. For each modification, determine whether the original theorem survives, and derive the strongest expression supported by the new model.

Investigation 1. Remove identical distribution but retain independence

Suppose $X_1, \dots, X_n$ are independent and satisfy $0 \leq X_i \leq 1$, but their means may differ: $\mathbb{E}[X_i] = \mu_i$. Define

$$\theta_n = \frac{1}{n} \sum_{i=1}^n \mu_i, \quad \hat{\theta}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

Does the Week 1 Hoeffding guarantee still hold? If so, derive its expression and identify its estimand. If the operational target is still the historical value $p_0 = 0.05$, what additional quantity must be controlled?

Working space

Thinking investigation. Replace independence by cluster dependence

An exact expression is recovered by identifying the independent units

Partition the $n = mb$ requests into m clusters of size b . Let $Y_1, \dots, Y_m$ be i.i.d. Bernoulli(p_0) and define

$$X_{j,k} = Y_j, \quad j = 1, \dots, m, \quad k = 1, \dots, b.$$

Outcomes inside a cluster are identical, while different clusters are independent. The empirical timeout rate becomes

$$\hat{p} = \frac{1}{mb} \sum_{j=1}^m \sum_{k=1}^b X_{j,k} = \frac{1}{m} \sum_{j=1}^m Y_j.$$

Investigation 2. Determine the valid theorem and the exact tail

Explain why Hoeffding with nominal sample size $n = mb$ is invalid, but a Hoeffding bound with m independent clusters is valid. Then derive an exact analytic expression for $\mathbb{P}(\hat{p} \geq t)$.

Specialize the expression to $n = 200$, cluster size $b = 10$, $m = 20$, $p_0 = 0.05$, and $t = 0.20$. Finally, recover the perfectly dependent common-state model by setting $m = 1$.

Working space

Thinking investigation. General dependence through covariance

An analytic upper bound may exist even when the exact tail is not identifiable

Suppose the X_i have common mean p_0 , common variance $\sigma^2 = p_0(1 - p_0)$, and common pairwise correlation ρ for $i \neq j$. Assume $0 \leq \rho \leq 1$. Then

$$\text{Cov}(X_i, X_j) = \rho\sigma^2.$$

Investigation 3. Derive the general variance expression

Starting from

$$\text{Var}(\hat{p}) = \frac{1}{n^2} \left(\sum_{i=1}^n \text{Var}(X_i) + 2 \sum_{i < j} \text{Cov}(X_i, X_j) \right),$$

derive $\text{Var}(\hat{p})$ as a function of n , σ^2 , and ρ . Use Chebyshev's inequality to obtain an analytic deviation bound. What happens as n tends to infinity when $\rho > 0$? Can the exact tail probability be recovered from pairwise correlation alone?

Working space

Synthesis. Return to the case and state the general lesson

Compare the four models considered in this investigation: i.i.d. Bernoulli, independent non-identical bounded observations, independent clusters with within-cluster dependence, and a covariance-specified model. For each, state what theorem or analytic expression is justified. Then give the QuickCart incident commander one immediate action and one additional data-collection step that would increase the amount of independent evidence.

Working space